EXP: 8 ACTIVITY DIAGRAM

AIM: To design and develop an ACTIVITY DIAGRAM for the system "PRECISION DISASTER RECOVERY (PDR-SYNC)" to represent the workflow and control flow of activities.

THEORY:

An ACTIVITY DIAGRAM is a UML behavioral diagram used to represent the flow of activities in a system. It shows the sequence of actions, decision points, and parallel processes.

It is mainly used for:

- Modeling workflow
- Representing business logic
- Understanding system behavior
- Process analysis

COMPONENTS OF ACTIVITY DIAGRAM:

INITIAL NODE: Represents the starting point of the process

ACTIVITY / ACTION: Represents a task performed in the system

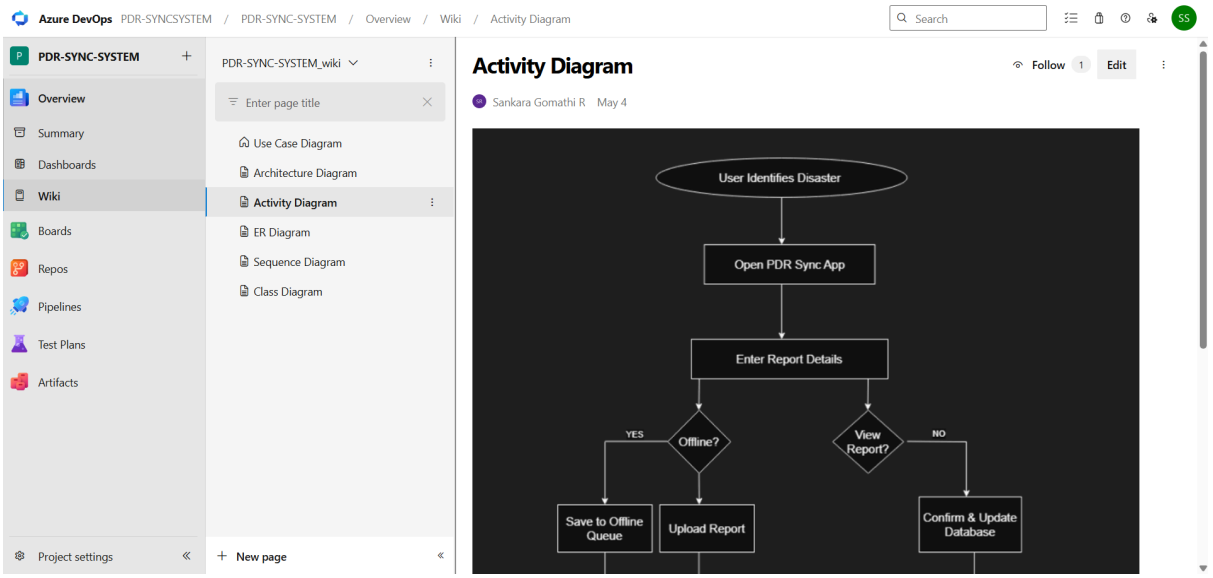
DECISION NODE: Represents a condition-based branching (YES/NO)

CONTROL FLOW: Shows the direction of execution

FINAL NODE: Represents the end of the process

NOTATIONS USED:

NOTATION	DESCRIPTION
Filled Circle	Initial Node
Rounded Rectangle	Activity
Diamond	Decision Node
Arrow	Control Flow
Double Circle	Final Node

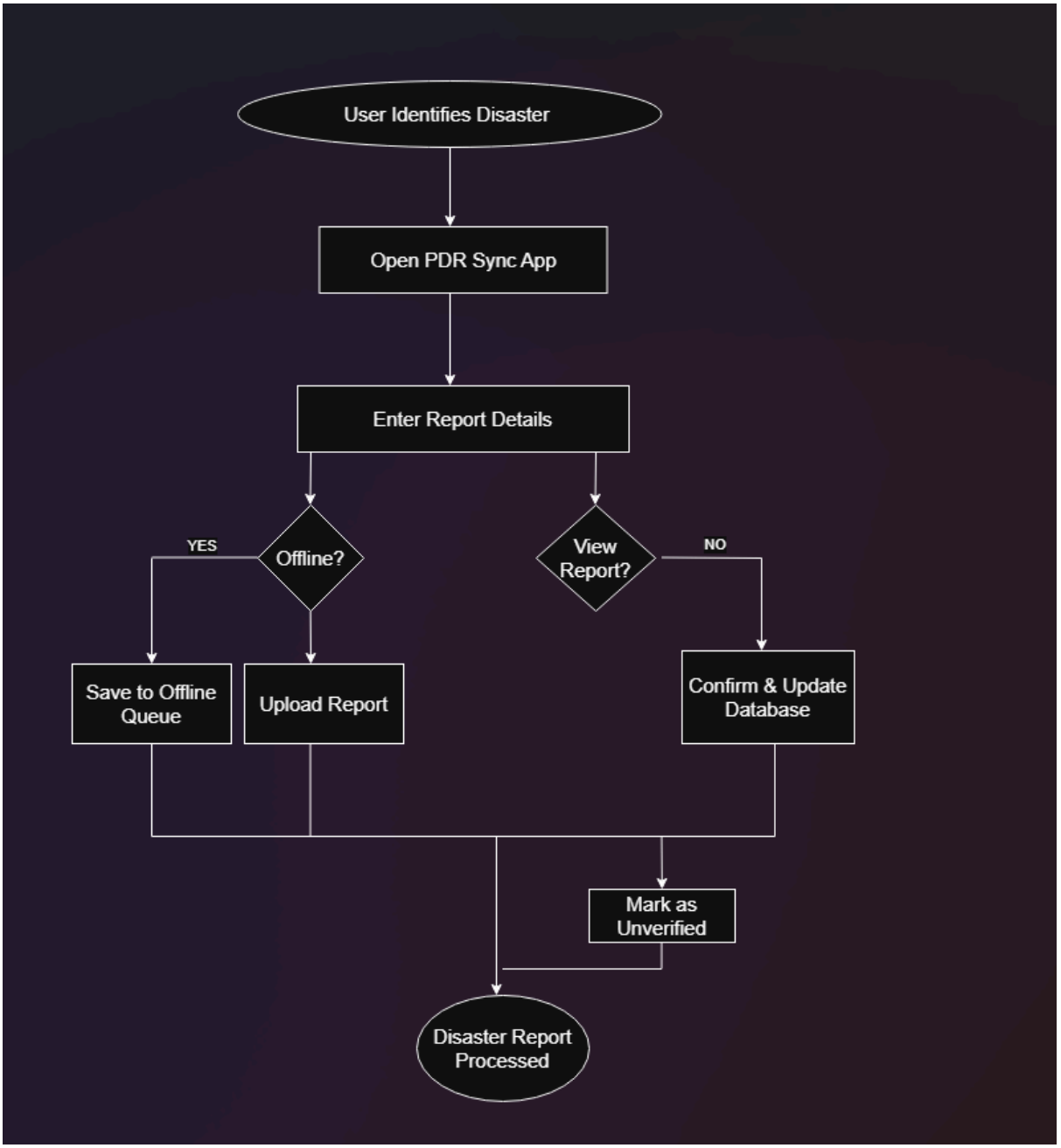


PROCEDURE:

1. Identify the workflow to be modeled
   (Disaster report submission and processing)
2. List all activities involved
3. Identify decision points (conditions)
4. Arrange activities in sequence
5. Connect activities using control flow arrows
6. Add start and end nodes
7. Validate diagram based on system logic

STEPS TO CREATE ACTIVITY DIAGRAM IN AZURE:

1. Login to Azure Portal
2. Open Azure DevOps Project
3. Navigate to Repos or Wiki
4. Use diagram tools such as draw.io
5. Create activity flow using UML shapes
6. Add decision nodes and conditions
7. Export diagram as image
8. Upload diagram to Azure DevOps



EXPLANATION OF ACTIVITY DIAGRAM:

The activity diagram represents the workflow of disaster report processing.

1. The user identifies a disaster
2. The PDR Sync application is opened
3. The user enters report details
4. The system checks whether the device is offline
    - If YES:
        - The report is saved in the offline queue
    - If NO:
        - The report is uploaded to the server

5. The system checks whether the user wants to view the report
    - If NO:
        - The database is updated
6. The report is marked as unverified
7. The disaster report is processed successfully

This workflow ensures:

- Offline-first functionality
- Reliable data handling
- Proper report processing

RESULT:

Thus, the ACTIVITY DIAGRAM for the system “PRECISION DISASTER RECOVERY (PDR-SYNC)” was successfully designed, representing the workflow and control flow of the system.